Tutorial

# Linear Algebra: Homework 4

Proofs: 16, 20, 21, 23, 24

True/false: 10(g), 17(a), 17(c), 17(f)

## Problem 10(g)

Let $V, W$ have ordered finite bases $\alpha, \beta$, and let $T : V \to W$ be linear.

**Original 10(g): label as true or false**

$$T = L_A \text{ for some matrix } A.$$

**False** $T$ acts on $V$, while $L_A : \mathbb{F}^n \to \mathbb{F}^m$ acts on coordinate columns.

$$A = [T]_\alpha^\beta, \qquad \boxed{[T(v)]_\beta = A[v]_\alpha}.$$

Here $n = \dim V$ and $m = \dim W$.

**Theorem 2.15(d)** If $S : \mathbb{F}^n \to \mathbb{F}^m$ is linear, then $S = L_C$ for a unique $m \times n$ matrix $C$.

Friedberg, Insel, Spence, Linear Algebra, 4th ed., Theorem 2.15(d), p. 93.

## Problem 10(g)

Let

$$V = \{f_{a,b} : \mathbb{R}^2 \to \mathbb{R} \mid f_{a,b}(x,y) = ax + by,\ a, b \in \mathbb{R}\}.$$

A basis is $\alpha = (\ell_1, \ell_2)$, where $\ell_1(x,y) = x$ and $\ell_2(x,y) = y$.

Define $T : V \to V$ by $T(f_{a,b}) = f_{a,-b}$. Then

$$A = [T]_\alpha^\alpha = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \qquad \dim V = 2.$$

For example,

$$T(f_{1,2}) = f_{1,-2}, \qquad L_A \begin{pmatrix} 1 \\ 2 \end{pmatrix} = \begin{pmatrix} 1 \\ -2 \end{pmatrix}.$$

$T$ acts on linear functions. $L_A$ acts on their coefficient columns.

For $S : \mathbb{R}^2 \to \mathbb{R}^2$ defined by $S(a,b) = (a,-b)$, $\boxed{S = L_A}$.

# Problem 17(a)

Let $V, W$ have ordered finite bases $\alpha, \beta$, and let $T : V \to W$ be linear.

**Original 17(a): label as true or false**

$$\left([T]_\alpha^\beta\right)^{-1} = [T^{-1}]_\alpha^\beta.$$

**False** The inverse reverses the input and output bases:

$$\boxed{\left([T]_\alpha^\beta\right)^{-1} = [T^{-1}]_\beta^\alpha}.$$

For $T = I$ on $\mathbb{R}^2$, take $\alpha = (e_1, e_2)$ and $\beta = (e_1, 2e_2)$. Then

$$A = [T]_\alpha^\beta = \operatorname{diag}(1, \tfrac{1}{2}), \qquad A^{-1} = \operatorname{diag}(1, 2) \neq [T^{-1}]_\alpha^\beta = A.$$

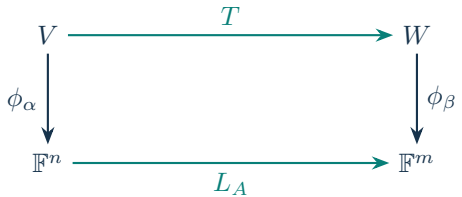
## Problem 17(c)

Let $V, W$ have ordered finite bases $\alpha, \beta$, and let $T : V \to W$ be linear.

**Original 17(c): label as true or false**

$$T = L_A, \qquad \text{where } A = [T]_\alpha^\beta.$$

**False** Define $\phi_\alpha(v) = [v]_\alpha$ and $\phi_\beta(w) = [w]_\beta$.




$$\boxed{T = \phi_\beta^{-1} \circ L_A \circ \phi_\alpha}.$$

## Problem 17(f)

Let $A$ and $B$ be matrices.

**Original 17(f): label as true or false**

$$AB = I \text{ implies that } A \text{ and } B \text{ are invertible.}$$

**False** The factors need not be square. Take

$$A = \begin{pmatrix} 1 & 0 \end{pmatrix} \in M_{1\times 2}(\mathbb{F}), \qquad B = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \in M_{2\times 1}(\mathbb{F}).$$

Then

$$AB = (1) = I_1, \qquad BA = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \neq I_2.$$

Neither rectangular factor has a two-sided inverse.

## Problem 16(a)

Let $V$ be a finite-dimensional vector space, and let $T : V \to V$ be linear.

**(a)** If $\operatorname{rank}(T) = \operatorname{rank}(T^2)$, prove that

$$\mathrm{R}(T) \cap \mathrm{N}(T) = \{0\}.$$

Deduce that

$$V = \mathrm{R}(T) \oplus \mathrm{N}(T)$$

(see the exercises of Section 1.3).

# Problem 16(a)

Let $V$ be a finite-dimensional vector space, and let $T : V \to V$ be linear. If $\operatorname{rank}(T) = \operatorname{rank}(T^2)$, prove that $\mathrm{R}(T) \cap \mathrm{N}(T) = \{0\}$. Deduce that $V = \mathrm{R}(T) \oplus \mathrm{N}(T)$ (see the exercises of Section 1.3).

First,

$$T(v) = 0 \Longrightarrow T^2(v) = 0, \qquad \mathrm{N}(T) \subseteq \mathrm{N}(T^2).$$

Rank–nullity and the rank hypothesis give

$$\begin{aligned} \dim \mathrm{N}(T) &= \dim V - \operatorname{rank} T, \\ \dim \mathrm{N}(T^2) &= \dim V - \operatorname{rank} T^2 = \dim \mathrm{N}(T). \end{aligned}$$

Nested subspaces of equal finite dimension are equal:

$$\boxed{\mathrm{N}(T) = \mathrm{N}(T^2)}.$$

**Common error.** Claiming $\mathrm{N}(T) = \mathrm{N}(T^2)$ without showing inclusion and equal dimensions.

## Problem 16(a)

Let $V$ be a finite-dimensional vector space, and let $T : V \to V$ be linear. If $\operatorname{rank}(T) = \operatorname{rank}(T^2)$, prove that $\mathrm{R}(T) \cap \mathrm{N}(T) = \{0\}$. Deduce that $V = \mathrm{R}(T) \oplus \mathrm{N}(T)$ (see the exercises of Section 1.3).

Take $v \in \mathrm{R}(T) \cap \mathrm{N}(T)$ and write $v = T(u)$. Then

$$T^2(u) = T(v) = 0.$$

Since $\mathrm{N}(T^2) = \mathrm{N}(T)$, we have $T(u) = 0$, hence $v = 0$. Thus $\mathrm{R}(T) \cap \mathrm{N}(T) = \{0\}$.

By the dimension formula and rank–nullity,

$$\dim\big(\mathrm{R}(T) + \mathrm{N}(T)\big) = \dim \mathrm{R}(T) + \dim \mathrm{N}(T) = \dim V.$$

The sum is a subspace of $V$ with full dimension, so

$$\boxed{V = \mathrm{R}(T) \oplus \mathrm{N}(T)}.$$

**Common error.** Stopping after $\mathrm{R}(T) \cap \mathrm{N}(T) = \{0\}$. The sum must also have dimension $\dim V$.

## Problem 16(b)

Let $V$ be a finite-dimensional vector space, and let $T : V \to V$ be linear.

**(b)** Prove that

$$V = \mathrm{R}(T^k) \oplus \mathrm{N}(T^k)$$

for some positive integer $k$.

# Problem 16(b)

Let $V$ be a finite-dimensional vector space, and let $T : V \to V$ be linear. Prove that $V = \mathrm{R}(T^k) \oplus \mathrm{N}(T^k)$ for some positive integer $k$.

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The ranges form a descending chain:

$$V \supseteq \mathrm{R}(T) \supseteq \mathrm{R}(T^2) \supseteq \cdots .$$

Their dimensions are nonincreasing nonnegative integers, so some $k \geq 1$ has equal adjacent ranks. Nestedness then gives

$$\mathrm{R}(T^k) = \mathrm{R}(T^{k+1}).$$

Applying $T$ repeatedly propagates the equality:

$$\mathrm{R}(T^k) = \mathrm{R}(T^{k+j}) \quad (j \geq 0).$$

In particular, $\boxed{\operatorname{rank} T^k = \operatorname{rank} T^{2k}}$.

**Common error.** Applying (a) to $T^k$ immediately after finding equal adjacent ranks, without justifying $\operatorname{rank} T^k = \operatorname{rank} T^{2k}$.

# Problem 16(b)

Let $V$ be a finite-dimensional vector space, and let $T : V \to V$ be linear. Prove that $V = \mathrm{R}(T^k) \oplus \mathrm{N}(T^k)$ for some positive integer $k$.

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Define $S = T^k : V \to V$. Then

$$S^2 = T^{2k}, \qquad \operatorname{rank} S = \operatorname{rank} S^2.$$

Part (a), applied to $S$, gives

$$\boxed{\mathrm{N}(S) \cap \mathrm{R}(S) = \{0\}}.$$

Rank–nullity gives $\dim \mathrm{R}(S) + \dim \mathrm{N}(S) = \dim V$. Hence

$$\boxed{V = \mathrm{R}(S) \oplus \mathrm{N}(S) = \mathrm{R}(T^k) \oplus \mathrm{N}(T^k)}.$$

## Problem 20

Let $A$ and $B$ be $n \times n$ matrices such that $AB$ is invertible. Prove that $A$ and $B$ are invertible.

Give an example to show that arbitrary matrices $A$ and $B$ need not be invertible if $AB$ is invertible.

## Problem 20

Let $A$ and $B$ be $n \times n$ matrices such that $AB$ is invertible. Prove that $A$ and $B$ are invertible. Give an example to show that arbitrary matrices $A$ and $B$ need not be invertible if $AB$ is invertible.

**$B$ is injective**

$$Bx = 0 \Longrightarrow ABx = 0 \Longrightarrow x = 0,$$

because $AB$ is invertible.

**$A$ is surjective** For every $y \in \mathbb{F}^n$, choose $x$ with

$$y = ABx = A(Bx).$$

Thus $y$ lies in the range of $A$.

Both factors act on $\mathbb{F}^n$. In finite equal dimensions, injectivity and surjectivity are equivalent. Hence both factors are invertible.

## Problem 20

Let $A$ and $B$ be $n \times n$ matrices such that $AB$ is invertible. Prove that $A$ and $B$ are invertible. Give an example to show that arbitrary matrices $A$ and $B$ need not be invertible if $AB$ is invertible.

Take

$$A = \begin{pmatrix} 1 & 0 \end{pmatrix} \in M_{1\times 2}(\mathbb{F}), \qquad B = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \in M_{2\times 1}(\mathbb{F}).$$

Then

$$AB = (1), \qquad BA = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}.$$

$AB$ is invertible, while neither rectangular factor has a two-sided inverse. The reversed product has a different size and is singular.

## Problem 21

For an arbitrary $n \times n$ matrix $A \in M_{n\times n}(\mathbb{R})$ with all entries smaller than $1$ for their absolute values, can we have

$$\lim_{m\to\infty} A^m = 0?$$

Justify your answer.

## Problem 21

For an arbitrary $n \times n$ matrix $A \in M_{n\times n}(\mathbb{R})$ with all entries smaller than $1$ for their absolute values, can we have $\lim_{m\to\infty} A^m = 0$? Justify your answer.

**A zero limit is possible** For $A = \frac{1}{2} I_n$,

$$A^m = 2^{-m} I_n \longrightarrow 0.$$

**The entry bound does not guarantee a zero limit**

$$H = \frac{1}{2}\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}, \qquad H^2 = H, \qquad H^m = H \neq 0 \quad (m \geq 1).$$

Every entry of $H$ has absolute value less than $1$.

An existence reading has answer yes. A guarantee for every matrix with the entry bound is false.

## Problem 23

Prove

$$\operatorname{rank}(L_A) + \operatorname{rank}(L_B) - n \leq \operatorname{rank}(L_{AB}) \leq \min\{\operatorname{rank}(L_A), \operatorname{rank}(L_B)\},$$

where $A$ and $B$ are $n \times n$ matrices.

**Original hint** Use the dimension inequality in Problem 22.

For a subspace $U$ of a finite-dimensional vector space $V$ and a linear map $T : V \to V$,

$$\boxed{\dim U - \dim \mathrm{N}(T) \leq \dim T(U) \leq \dim U}.$$

## Problem 23

Prove $\operatorname{rank}(L_A) + \operatorname{rank}(L_B) - n \leq \operatorname{rank}(L_{AB}) \leq \min\{\operatorname{rank}(L_A), \operatorname{rank}(L_B)\}$, where $A$ and $B$ are $n \times n$ matrices. Hint: Use the dimension inequality in Problem 22.

Since $L_{AB} = L_A \circ L_B$,

$$\mathrm{R}(L_{AB}) \subseteq \mathrm{R}(L_A) \quad \Longrightarrow \quad \operatorname{rank}(L_{AB}) \leq \operatorname{rank}(L_A).$$

Put $U = \mathrm{R}(L_B)$. Then $\mathrm{R}(L_{AB}) = L_A(U)$, so

$$\operatorname{rank}(L_{AB}) = \dim L_A(U) \leq \dim U = \operatorname{rank}(L_B).$$

Therefore

$$\boxed{\operatorname{rank}(L_{AB}) \leq \min\{\operatorname{rank}(L_A), \operatorname{rank}(L_B)\}}.$$

## Problem 23

Prove $\operatorname{rank}(L_A) + \operatorname{rank}(L_B) - n \le \operatorname{rank}(L_{AB}) \le \min\{\operatorname{rank}(L_A), \operatorname{rank}(L_B)\}$, where $A$ and $B$ are $n \times n$ matrices. Hint: Use the dimension inequality in Problem 22.

Problem 22 gives

$$\dim U - \dim \mathrm{N}(T) \le \dim T(U) \le \dim U.$$

Take $V = \mathbb{F}^n$, $U = \mathrm{R}(L_B)$, and $T = L_A$. Then $T(U) = \mathrm{R}(L_{AB})$,

$$\dim U = \operatorname{rank}(L_B), \qquad \dim \mathrm{N}(L_A) = n - \operatorname{rank}(L_A).$$

Substituting into the left inequality,

$$\begin{aligned} \operatorname{rank}(L_{AB}) &\ge \operatorname{rank}(L_B) - \big(n - \operatorname{rank}(L_A)\big) \\ &= \operatorname{rank}(L_A) + \operatorname{rank}(L_B) - n. \end{aligned}$$

**Common error.** Taking $U = \mathrm{R}(L_A)$ and $T = L_B$ for $AB$. That choice gives $T(U) = \mathrm{R}(L_{BA})$.

## Problem 23

Prove $\operatorname{rank}(L_A) + \operatorname{rank}(L_B) - n \leq \operatorname{rank}(L_{AB}) \leq \min\{\operatorname{rank}(L_A), \operatorname{rank}(L_B)\}$, where $A$ and $B$ are $n \times n$ matrices. Hint: Use the dimension inequality in Problem 22.

Take square matrices

$$A = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \qquad B = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}.$$

Then $AB = B$ and $BA = O$, so

$$\boxed{\operatorname{rank}(L_{AB}) = 1, \qquad \operatorname{rank}(L_{BA}) = 0}.$$

Both factors have rank $1$. Problem 23 bounds each product rank between $0$ and $1$, so both values satisfy the inequality.

**Common error.** Assuming $\mathrm{R}(L_{AB}) = \mathrm{R}(L_{BA})$ or $\operatorname{rank}(L_{AB}) = \operatorname{rank}(L_{BA})$.

## Problem 24

Prove

$$\operatorname{rank}(L_{A+B}) \leq \operatorname{rank}(L_A) + \operatorname{rank}(L_B),$$

where $A$ and $B$ are $n \times m$ matrices.

## Problem 24

Prove $\operatorname{rank}(L_{A+B}) \le \operatorname{rank}(L_A) + \operatorname{rank}(L_B)$, where $A$ and $B$ are $n \times m$ matrices.

For $x \in \mathbb{F}^m$, $(A+B)x = Ax + Bx$. Therefore

$$\mathrm{R}(L_{A+B}) \subseteq \mathrm{R}(L_A) + \mathrm{R}(L_B).$$

Taking dimensions,

$$\begin{aligned} \operatorname{rank}(L_{A+B}) &\le \dim\big(\mathrm{R}(L_A) + \mathrm{R}(L_B)\big) \\ &= \operatorname{rank}(L_A) + \operatorname{rank}(L_B) - \dim\big(\mathrm{R}(L_A) \cap \mathrm{R}(L_B)\big) \\ &\le \operatorname{rank}(L_A) + \operatorname{rank}(L_B). \end{aligned}$$

The inclusion can be strict: $A = I_n$, $B = -I_n$ give $\mathrm{R}(L_{A+B}) = \{0\}$ but $\mathrm{R}(L_A) + \mathrm{R}(L_B) = \mathbb{F}^n$.

**Common error.** Writing $\mathrm{R}(L_{A+B}) = \mathrm{R}(L_A) + \mathrm{R}(L_B)$. The right side allows independent inputs; $(A+B)x$ uses the same input.

## Extension

Let $A$ and $B$ be $n \times n$ matrices over $\mathbb{F}$. Prove that

$$\boxed{\operatorname{rank}(L_{AB}) = \operatorname{rank}(L_B) - \dim\bigl(\mathrm{R}(L_B) \cap \mathrm{N}(L_A)\bigr)}.$$

**Hint** Apply rank–nullity to the restriction of $L_A$ to $\mathrm{R}(L_B)$.

## Extension

Let $A$ and $B$ be $n \times n$ matrices over $\mathbb{F}$. Prove that $\operatorname{rank}(L_{AB}) = \operatorname{rank}(L_B) - \dim\big(\mathrm{R}(L_B) \cap \mathrm{N}(L_A)\big)$.

Put $U = \mathrm{R}(L_B)$ and consider the restriction

$$S = L_A|_U : U \to \mathbb{F}^n, \qquad S(y) = Ay.$$

Its range and kernel are

$$\begin{aligned} \mathrm{R}(S) &= L_A(U) = \mathrm{R}(L_{AB}), \\ \mathrm{N}(S) &= \{y \in U : Ay = 0\} = U \cap \mathrm{N}(L_A). \end{aligned}$$

Rank–nullity for $S$, whose domain is $U$, gives

$$\operatorname{rank}(L_B) = \dim U = \dim \mathrm{R}(S) + \dim \mathrm{N}(S).$$

Therefore

$$\boxed{\operatorname{rank}(L_{AB}) = \operatorname{rank}(L_B) - \dim\big(\mathrm{R}(L_B) \cap \mathrm{N}(L_A)\big)}.$$